

FIELDCELL · REQUIREMENTS

FieldCell requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/fieldcell		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

FieldCell requirements

These are first-pass requirements for the concept. Targets are proposals for review and will be checked by calculation at TRL 3. The estimate column comes from the first-order numbers in FCL-PRC-001.

Table 1. Requirements.

ID	Requirement	Target	Concept estimate	Verification (TRL 3 or later)
R1	Store enough energy for a day of field use	1.2 kWh nominal or more; 1.0 kWh usable or more	1.28 kWh nominal, about 1.15 kWh usable at 90 % depth of discharge. Met	Datasheet and energy calculation
R2	Supply AC and DC loads	1,000 W continuous pure sine AC, 2,000 W surge; DC outlets 300 W or more in total (USB-C PD and 12 V)	1 kW inverter; about 440 W DC (2 x 100 W USB-C, 240 W at 12 V). Met by selection	Datasheets; current and cable calculation
R3	Recharge from its own PV	1.1 kWh/day or more into the battery at 4 peak sun hours; full recharge from 10 % in one day at 5 peak sun hours with no load	1.12 kWh/day at 4 h (thin); full recharge in about 0.8 day at 5 h. Met, thin at 4 h	PV yield calculation with stated derating
R4	Carry a reference load through a day without sun	Reference load of 0.9 kWh/day at the outlets for 1 day or more with no sun; energy-neutral at 4 peak sun hours or more	About 1.1 days; about 9 % surplus at 4 h. Met, thin	Load profile and energy balance
R5	Deploy fast with one person	From arrival to power available in 10 min or less, one person, no tools; stow in 10 min or less	About 6 min (step estimate). Met	Task analysis; later timed trial
R6	Movable by one person	Total mass 70 kg or less; vertical handle force 10 to 150 N (never negative) over a 5 degree pitch either way; heaviest removable module 25 kg or less	About 68.5 kg (thin); 17 to 52 N; battery pack about 12 kg. Met, mass margin thin	Mass budget and center of gravity calculation
R7	Cross rough ground	16 in (406 mm) wheels or larger; ground clearance 150 mm or more; climb a 150 mm step; pull force 150 N or less on a 10 % grade over gravel or grass; static lateral tip angle 25 degrees or more	406 mm wheels; about 190 mm clearance; pull about 110 to 135 N; tip about 32 degrees. Met, pull force thin	Geometry and force calculation
R8	Work in rain and dust	Electronics enclosure IP54 while operating; battery enclosure IP65; all cable entries through glands or sealed connectors	Design intent; ventilation of the electronics box against IP54 not yet resolved. Open	Design review; later spray test
R9	Stay safe across field temperatures	Operate at -10 to 45 °C ambient; BMS blocks charging below 0 °C cell temperature; inverter derating known at 45 °C	Design intent; thermal model not done. Open	Datasheets and thermal estimate
R10	Stay put in wind when deployed	Wings and cart stable to 15 m/s (54 km/h) with the four outrigger feet staked or ballasted; stow instruction above that	Unstaked wings may lift at about 9 m/s. Met only with stakes	Wind load calculation
R11	Be quiet	45 dB(A) or less at 1 m at 500 W load	Depends on inverter fan; unverified	Datasheet; later sound level measurement
R12	Be electrically safe and affordable	Every circuit fused at its source; GFCI (120 V) or 30 mA RCD (230 V) on all AC outlets; battery disconnect reachable without tools; no inlet or path to building wiring; parts cost \$1,500 or less	Design intent; parts about \$1,462 (indicative), about 2.5 % margin. Met, no contingency	Design review; priced BOM

■ Assumptions

- Peak sun hours of 4 to 5 per day (clear-season mid-latitude sites); a cloudy week is not covered by R3 or R4.
- The reference load in R4 is defined in FCL-PRC-001: lighting, radio and phone charging, a laptop and router, and short power-tool bursts.
- Usable battery capacity is 90 % of nominal, for LiFePO4 cells with a conservative BMS.
- "Arrival" in R5 means the cart has been wheeled to the chosen spot; choosing the spot is not timed.